MANOJ SRIVATSAV

B.Tech Student in Computer Science and Artificial Intelligence

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EXPERIENCE

Computer Vision Intern

FitFlix

June 2024 - August 2024   Bengaluru, India

- Worked on a real-time posture correction system for fitness applications using Python, OpenCV, and pose-estimation land- marks.
- Implemented live angle computation for hips, knees, and ankles using custom functions (find angle, get landmark features)
- Developed a full processing pipeline that tracked inactivity, counted correct/incorrect squats, displayed visual overlays, and produced real-time feedback using text overlays, dotted lines, and arc-based joint angle visualization.

PROJECTS

Emotional Detection Using Multi-Modal Analysis

Python, OpenCV, Dlib, Librosa, TensorFlow

- Built a multi-modal system combining facial expression and speech analysis for emotion detection.
- Implemented deep learning pipelines for FER and audio-based emotion recognition.
- Designed a real-time detection module demonstrating applications in HCI and mental health monitoring.

Quantum Maze Solver

Qiskit, Python, Quantum Computing

- Developed a quantum maze solver using superposition and Grover's algorithm for optimal pathfinding.
- Designed dynamic maze visualization to compare classical and quantum approaches.
- Analyzed and improved processes. Performed fatigue, beam, static torque, gear fatigue, deflection, hot/cold flow, and axle chatter tests.

Green Video Streaming Optimization Framework

Python, AWS EC2, Random Forest, Cloud Computing

- Developed an energy-aware video transcoding system that optimizes complexity, energy usage, and CO2 emissions for cloud-based streaming.
- Implemented content-aware analysis and Random Forest models to predict CPU load, encoding time, and quality metrics for adaptive encoding.
- Deployed on AWS EC2, achieving up to 22.6 percent lower CPU usage, 30percent reduced energy consumption, and 16.7 percent lower carbon emissions.

KEY ACHIEVEMENTS

Heat Transfer Prediction using Neural Networks
Predicting heat transfer coefficients in a double-pipe heat exchanger using data-driven neural network models.

Post-Earthquake Damage Assessment Using Siamese U-Net with Triplet Attention
Developed a Siamese U-Net damage segmentation model with TE23D data that surpassed key baselines on all major metrics.

STRENGTHS

Hard-working | Disciplined | Experienced Teacher & Motivator | Focused | Reliable

LANGUAGES

English	●●●●●
Hindi	●●●●○
Telugu	●●●●●

SKILLS

Technical: Neural Networks, Python, Java, JavaScript, DSA, Autopsy Regression Models Frameworks: React, Node.js, Flask, Docker, Wireshark, Tensorflow, OpenCv, Keras, Pytorch

CERTIFICATIONS

AWS Academy - AWS Cloud Foundation

- IBM Machine Learning